Deval Mehta, PhD

Research Fellow in Artificial Intelligence for Healthcare and Digital Health

Founding Member, AIM for Health Lab, Monash University

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PROFILE

I am an interdisciplinary research leader with academic, clinical, and industry experience in research and development of Translational AI systems for healthcare. My expertise spans multimodal learning, foundation AI models, LLMs, and clinician-in-the-loop design for healthcare problems. I have developed and led a team of PhD students, interns, and research assistants to translate AI models into clinical practice across neurodegenerative diseases (brain), dermatology (skin), and ophthalmology (eye). Through strong industry collaborations, my work has advanced from lab to practice and is published in top AI and clinical venues. I have attracted more than AUD $1.7 million in funding as a Chief Investigator across several AI for Healthcare projects. My teaching spans applied AI, ethical considerations, and curriculum co-design for future-ready AI education, especially for the healthcare domain.

Education

PhD, School of Electrical and Electronics Engineering — 2014-2019
Nanyang Technological University, Singapore

BTech, Department of Electrical and Electronics Engineering — 2010-2014
National Institute of Technology, Trichy, India

Work and research Experience

Research Fellow	Faculty of IT, Monash University	May 2021 - present
Honorary Research Fellow	Alfred Health	Mar 2022 - present
Founding Member	AIM for Health Lab, Monash University	July 2023 - present
Steering Committee Member	Victorian Heart Institute	July 2025 - present
Research Scientist	IBM Research Labs Australia	Feb 2020 - Apr 2021
Senior R&D Engineer	Panasonic R&D Center Singapore	Apr 2019 - Nov 2019
R&D Engineer	Panasonic R&D Center Singapore	Jul 2018 - Mar 2019

Ongoing Key Projects and Collaborations

The below collaborations are initiated and led by me at Faculty of IT, Monash University:

- AI for Neurology - I initiated this collaboration with Prof. Patrick Kwan for developing Foundation AI models and AI Agents for neurology disease management [2021 - present].
- IIT-Bombay - AIM for Health Lab Joint Venture - I initiated this collaboration with Prof. Kshitij Jadhav from KCDH, IIT-B to develop Generative AI and Foundation Models for skin [2024-present].
- Medical Knowledge Base Generation - I initiated this collaboration Prof. Karin Verspoor and Assoc. Prof. Sonika Tyagi at RMIT to develop a comprehensive knowledge base using graph networks and LLMs for different healthcare areas like pregnancy and epilepsy [2025-present].
- Patient and Clinician Centered Design - I initiated this collaboration with Epilepsy Foundation and Dr. Joe Liu (Dept. HCC) for developing AI systems incorporating patient and clinician feedback.

Selected Publications

19 Q1 and CORE A*/A author publications across technical AI conferences and journals, and clinical translation journals. For full list, please refer to Google Scholar Profile. Some of the highlighted works are described below (* refers to corresponding and/or first author):

- Yiwen, J., D. Mehta*, Yan, S., Ge, Z., "WISE: Weak-Supervision-Guided Step-by-Step Explanations for Multimodal LLMs in Image Classification", EMNLP (Main), August 2025. (Top conference in Natural Language Processing). **Supervising author of student**.
- Yiwen, J., D. Mehta*, W. Fei, Ge, Z. ,"Enhancing Interpretable Image Classification Through LLM Agents and Conditional Concept Bottleneck Models", ACL (Main), June 2025. (Top conference in Natural Language Processing). **Supervising author of student**.
- Pham, K., D. Mehta*, Verspoor, K., Kwan, P., Ge, Z. ,"Knowledge Tree Driven Contextualized Instruction Tuning of Foundation Models for Epilepsy Drug Recommendation", MICCAI, June 2025. (Top conference in Medical AI). **Supervising author of student**. (AI for Epilepsy)
- Yang, H., George, Y., D. Mehta, Lin, L., Yang, D., Ge, Z. , "Multi-modal CT Perfusion-based Deep Learning for Predicting Stroke Lesion Outcomes in Complete and No Recanalization Scenarios", American Journal of Neuroradiology, June 2025. (Top Q1 journal in Neurology). (AI for Stroke).
- R. Kumar, R. Singhal, P. Kulkarni, D. Mehta, K. Jadhav ,"M3CoL: Harnessing Shared Relations via Multimodal Mixup Contrastive Learning for Multimodal Classification", Transactions on Machine Learning Research, May 2025. (Top Q1 journal in Machine Learning).
- D. Mehta*, Yiwen, J., Catherine, T., Mingguang, H., Kshitij, J., Ge, Z. ,"Interpretable Few-Shot Retinal Disease Diagnosis with Concept-Guided Prompting of Vision-Language Models", Information Processing in Medical Imaging, Feb 2025. (Top conference in Medical AI since 1969). **First and corresponding author**. (AI for Eye Disease)
- D. Mehta*, Primerio, C., Gal, Y., Mar, V., Soyer, P., Ge, Z. ,"Large Multi-task Vision AI Model for Dermatology: Overcoming Critical Challenges in Skin Lesion Diagnosis", Journal of the European Academy of Dermatology and Venereology, Jan 2025. (Top Q1 clinical journal in skin disease, acceptance rate: 10%). **First and corresponding author**. (AI for Skin)
- Zhang, X., D. Mehta, Zhu, C., Darby, D., Ge, Z., Helmut, B. ,"Adaptive Transformer Modelling of Density Function for Nonparametric Survival Analysis", Journal of Machine Learning, Jan. 2025. (Top Q1 journal in Machine Learning & AI). **Supervising author of student**. (AI for Multiple Sclerosis)
- D. Mehta*, Sivathamboo, S., Simpson, H., Kwan, P., Ge, Z. ,"Privacy-Preserving Early Detection of Epileptic Seizures in Videos", MICCAI, 2023. (Top conference in Medical Imaging)
  Top 15% paper early accept out of 2,500. **First and corresponding author**. (AI for Epilepsy)
- Asif, U., D. Mehta, Von Cavallar, S., Tang, J., Harrer, S. ,"DeepActsNet: A deep ensemble framework combining features from face, hands, and body for action recognition", Pattern Recognition, 139, 109484. 2023. (Top Q1 journal in AI).
- D. Mehta*, Gal, Y., Bowling, A., Bonnington, P., Ge, Z.,"Out-of-distribution detection for long-tailed and fine-grained skin lesion images", MICCAI, 2022.
  Top 15% paper early accept out of 2,000. **First and corresponding author**. (AI for Skin)
- D. Mehta*, Asif, U., Hao, T., Bilal, E., Von Cavallar, S., Harrer, S., Rogers, J., "Towards automated and marker-less Parkinson disease assessment: predicting UPDRS scores using sit-stand videos", CVPR (pp. 3841-3849) 2021. (Top conference in AI). **First and corresponding author**. (AI for Parkinson's)

Research Grants

» Awarded Grants: Sum total - AUD $1,755,036

- D. Mehta, "Advancing Cross-national Digital Health Collaboration: Improving Skin Care Through AI-powered System for Rural Communities in Australia and India", AUD $13,500, **Australian Academy of Science Australia–India Strategic Research Fund** 2026.
- Kwan, P., D. Mehta, Ge. Z., "MRI Foundation Model for Neurology Imaging and Clinical Translation", AUD $96,700 [32K A100 GPU hours], **NVIDIA Academic Grant** 2025-26.
- Ge, Z., D. Mehta, Kwan, P., Meng, L., "Neuroimaging Foundation Model Infrastructure for Research and Clinical Translation", AUD $943,325, **National Imaging Facility (NIF) Infrastructure Grant** 2025-27.
- D. Mehta, Foster, E., Kwan, P., "A Digital Risk Stratification Tool to Reduce Repeat Seizure Presentations to Emergency Departments", AUD $50,000, **AI in Health Seed Grant Scheme** 2025-26.
- Liu, J., Haryanto, A., D. Mehta, Satriadi, K., "HandovAR: A Framework of AI-facilitated Handover System via Augmented Reality for ICU Nurses", AUD $30,000, **Faculty of IT Seed Grant** 2025-26.
- Ge, Z., George, Y., D. Mehta, "Collaboration with Optiscan Imaging", AUD $564,540, **Digital Health CRC-P** 2024-27.
- George, Y., D. Mehta, Kwan, P., "Multimodal Federated Learning for Epilepsy Care and Management", AUD $34,380, **Faculty of IT Seed Grant** 2024-25.
- Pei, X., Shin, W., D. Mehta, "Comprehending the challenges and opportunities faced by Australian elderly communities in medical AI adoption", AUD $19,291, **University of Melbourne Arts Collaborative Research Seed Fund** 2022-23.

» Applied Grants awaiting decision

- Tyagi, S., D. Mehta, Peleg, A., "Orchestrated Multi-Modal Agentic AI for Infection Prevention and Control in Residential Aged Care", **Amazon Web Services (AWS) Agentic AI** , Nov 2025.
- Sinclair, B., D. Mehta, O'Brien, T., Kaestner, E., "Deep Learning For Epilepsy Surgery", **MRFF EMCR**, July 2025.

Teaching Experience

- Guest Lecturer, "**Deep Learning**" (Postgraduate) at Monash - S2 2025.
  Key contributions: Teaching the evolution of sequence to sequence learning and transformers in natural language processing to computer vision domain.
- Guest Lecturer and Tutor, "**Foundations of Data Science**" (Undergrad) at Monash - S2 2024.
  Key contributions: Designed an engaging and interactive framework for demonstrating the pitfalls with standard performance metrics used for evaluating AI models.
- Tutor, "**Systems Analysis and Design**" (Postgraduate) at Monash - S1 2024.
  Key contributions: Led the coordination of tutor marking duties and redesigned assignments to align with course objectives.
- Tutor, "**Statistical Data Modeling**" (Postgraduate) at Monash - S2 2023.
  Key contributions: Designed new slides to simplify difficult statistical concepts like binomial distributions and confidence intervals for students.
- Guest Lecturer, "**Frontiers in Data Science**"(Undergrad) at Monash - 2023-25.

Key contributions: Designed innovative lectures to demonstrate real-world challenges and implementation of AI for healthcare analytics.

- Assisting in the conceptualisation and design of a new course on AI in healthcare. Providing input on curriculum structure, learning outcomes, and applied AI use-cases, in collaboration with Prof. Chris Bain.

Graduate Research Supervision

- Project Lead supervisor of 2 successfully graduated PhD students at Monash and 3 ongoing PhD students at Faculty of IT and Faculty of Engineering, Monash University.
- Associate supervisor of 2 ongoing medical PhD students at Faculty of Medical Nursing and Health Sciences (MNHS), Monash University.
- Associate supervisor of 1 ongoing PhD student and 1 ongoing Master Student at IIT-Bombay.
- Project Lead supervisor of successful completion of 5 Final Year Project and 2 Master by Minor Thesis students.

Patents

Commercial utility of my research is demonstrated through below intellectual property.

- Deval Mehta, Talha Ilyas, Zongyuan Ge, "Method and System for De-identified Motion Tracking", **Australia Provisional Patent** 2025905326, Nov 2025.
- Deval Mehta, Yiwen Jiang, Zongyuan Ge, Minguang He, Jason Sun, "Multimodal Data Collection And Diagnosis Of Rare Ocular Diseases", **US Patent App.** WO2024US59516, Jan 2024.
- Deval Mehta, Toan Ngyuen, Yaniv Gal, Zongyuan Ge, Paul Bonnington,"Out of Distribution Detection for Long-tailed and Fine-grained Datasets", **New Zealand Patent App.** 786708, Mar 2022.
- Umar Asif, Deval Mehta, Stefan von Cavallar, Stefan Harrer, Jianbin Tang, Tian Hao, Jeffrey L. Rogers, Erhan Bilal,"Privacy-preserving motion analysis", **US Patent App.** 17/249,132, Jan 2021.
- Tian Hao, Jeffrey L. Rogers, Umar Asif, Erhan Bilal, Deval Mehta, Stefan Harrer, Jianbin Tang, Stefan von Cavallar,"Assessing parkinson's disease symptoms", **US Patent App.** 17/249,412, Aug 2020.
- Deval Mehta, Shoushun Chen, and Kay Soon Low,"A High Accuracy Star Tracker Using Running Sequential Angular Match Technique", **Singapore Copyright**, 10201603223W, April 2016.

Keynote Talks

- "Medical Foundation Models for Retina, Skin, and Brain at Population Scale", NVIDIA AI Day Sydney, 2025.
- "Towards Out-of-Hospital Monitoring of Epilepsy Patients", Sharing About Healthcare Artificial Intelligence Regionally (SHARE), 2025.
- "Getting Artificial Intelligence Ready for Clinic: Revamping AI Models in Dermatology", IEEE Western Australia Joint Computational Intelligence / Robotics & Automation Talk, 2024.
- "Foundation Models for Skin Lesion Analysis", Melanoma Research Victoria Scientific Exchange Meeting 2023.

Academic Fellowships & Achievements

- Early Career Researcher Award, Australia-India Strategic Research Fund, Australian Academy of Science, 2026
- One of the top 3 Finalists for the Victoria Melanoma Young Researcher Award (Morgan Mansell Prize), 2023
- Selected as one of the top 200 early career scientist globally to attend the 10th Heidelberg Laureate Forum (HLF), 2023
- Awarded the HLF travel grant from Australian Academy of Science for attending the 10th HLF (Only 1 in Australia selected every year), 2023
- Awarded TOSHIBA Global Intern at Toshiba Devices & Storage Corporation Tokyo, Japan, 2018
- Nanyang Technological University Research Scholarship for PhD, 2014-2018
- DAAD WISE Scholarship Technical University of Munich, Germany, 2013

References

» Academic referees from computing domain:

- Karin Verspoor, PhD, Dr. | Dean, School of Computing Technologies
  RMIT University - Australia
  karin.verspoor@rmit.edu.au
- Chris Bain, PhD, Dr. | Prof. of Practice, Faculty of IT
  Monash University - Australia
  chris.a.bain@monash.edu
- Sonika Tyagi, PhD, Dr. | Associate Professor, School of Computing Technologies
  RMIT University - Australia
  sonika.tyagi@rmit.edu.au
- Stefan Harrer, PhD, Dr. | Director, AI4Science
  CSIRO - Australia
  stefan.harrer@data61.csiro.au

» Academic referees from medical domain:

- Patrick Kwan, MD, PhD, Dr. | Professor of Neurology, Faculty of MNHS
  Monash University - Australia
  patrick.kwan@monash.edu
- Victoria Mar, MD, PhD, Dr. | Director of Victoria Melanoma Service
  Alfred Health - Australia
  victoria.mar@monash.edu

» Industry referees:

- Yaniv Gal, PhD, Dr. | AI/ML Tech Lead, Orion Health
  Previous CTO of Molemap Pty. Ltd. - Australia
  yanivgal.work@gmail.com
- Jeff Dunkel | CEO, Optain Health Pty. Ltd.
  jeff@optainhealth.com